University of Alberta

Chemical and Materials Engineering

MAT E 374

Computational Methods in Materials Engineering

Midterm examination

20% of final grade

Friday, October 30, 2026
12:00–12:50 pm · 50 minutes

Name: _____ Student ID: _____

Instructions

- **Three questions worth 100 points**: Q1 (20), Q2 (40), and Q3 (40). The duration is **50 minutes**.
- **Open book.** You may consult printed materials, course materials, notebooks, assignment solutions, and Python/API documentation. Search engines may be used to find documentation or debugging information.
- **Generative AI and communication with other people are prohibited**, including AI assistance in searches, debugging, or writing code.
- **Permitted methods.** Unless a question specifies a method, you may use pen and paper, calculator-assisted calculations (including Python as a calculator), or Python code. Show the method, intermediate steps, and results.
- **Only answers on the answer sheet will be marked.** Include units and supporting calculations. If using Python, identify the method and functions used.
- **Upload any code through Canvas** for record keeping. If uploading fails, write the key code on the answer sheet. Code quality is not graded.
- **No official formula sheet is provided.** You may bring your own.
- **If a question is unclear,** state your interpretation or assumptions and proceed. Do not ask for clarification of examination questions. Ask for technical help if you cannot access the coding environment.

Q1 · Short answers (20 points)

Answer each part concisely.

(a) Step size and accuracy (10 points)

A program calculates a materials property using a parameter `step_size`. Reducing `step_size` from 0.1 to 0.01, and then to 0.001, reduces the relative error at each step. Should you directly use a step size of 10^{-10} to ensure a very accurate result? State **yes or no** and give **one** reason.

(b) Newton–Raphson (10 points)

When preparing for the MATE374 midterm, you ask an AI assistant about Newton–Raphson. The assistant claims that for $f(x) = ax + b$, with $a \neq 0$, Newton–Raphson reaches the root in **one iteration** from any initial guess. Is this statement correct? Give **one** reason.

Q2 · The Peng–Robinson equation of state (40 points)

The Peng–Robinson equation of state, developed at the University of Alberta and published in 1976, relates pressure P, temperature T, and molar volume V:

$$P = \frac{RT}{V - b} - \frac{a(T)}{V^2 + 2bV - b^2},$$

The parameters are

$$a(T) = 0.45724 \frac{R^2 T_c^2}{P_c} \left[1 + \kappa \left(1 - \sqrt{T/T_c}\right)\right]^2, \qquad b = 0.07780 \frac{RT_c}{P_c},$$

$$\kappa = 0.37464 + 1.54226\omega - 0.26992\omega^2.$$

For carbon dioxide, use $T_c = 304.1282$ K, $P_c = 73.7703$ bar, $\omega = 0.22394$, and $R = 0.0831446$ L bar mol^{-1} K^{-1}. Note that a depends on T.

(a) Residual function (10 points)

Write down the expression for the residual function $F(V; T, P) = 0$. You do not need to expand a, b, or κ.

(b) Numerical implementation (10 points)

Write `aT(T)` and `F(V, T, P)` in Python. Constants are provided in the Canvas quiz notebook. For calculated pressure minus specified pressure, check that

$$F(V = 1 \text{ L mol}^{-1};\ T = 400 \text{ K},\ P = 1 \text{ bar}) \approx 30.14464 \text{ bar}.$$

(c) Root finding (20 points)

Use your functions to find **all three roots with** $V > b$ at $T = 280$ K and $P = 40$ bar. Choose any Python root-finding method, using Figure 1 for brackets or initial guesses. State your method and report the roots in L mol^{-1} to at least three significant figures.

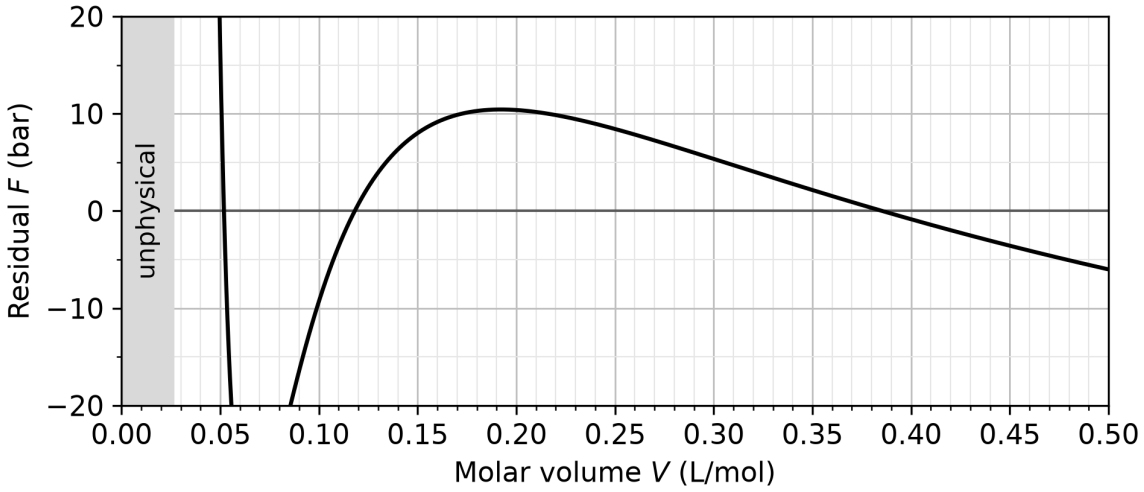


Figure 1: Residual at 280 K and 40 bar. The horizontal line marks zero, and the shaded band marks the unphysical region $V \leq b$.

Q3 · Mass-spectrometry measurements (40 points)

A mixture contains four compounds, A, B, C, and D, with unknown amounts x_A, x_B, x_C, x_D. A mass spectrometer measures the total intensity at four mass-to-charge ratios (m/z). Both intensities and amounts are unitless in this problem.

Each measured intensity is the sum of the four compounds' contributions, using the response coefficients in the table.

m/z	A	B	C	D	Measured intensity
27	4	1	3	2	21
28	8	2	4	2	34
30	0	7	2	9	29
44	0	2	9	0	31

For example, at $m/z = 27$, the four contributions give

$$4x_A + x_B + 3x_C + 2x_D = 21.$$

(a) Matrix formulation (10 points)

Express the measurements as $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{A}$ is the intensity coefficient matrix, $\mathbf{x} = (x_A, x_B, x_C, x_D)^T$ is the amount vector, and $\mathbf{b}$ is the measured intensity vector. Write $\mathbf{A}$, $\mathbf{x}$, and $\mathbf{b}$.

(b) Amounts and verification (15 points)

Solve for the amounts $\mathbf{x}$ by hand or with code. Verify your results by comparing $\mathbf{Ax}$ with $\mathbf{b}$.

(c) Correcting a measurement (10 points)

After checking the experimental setup, you find that a faulty sensor caused the peak at $m/z = 27$ to be recorded incorrectly. Its true intensity is 22, and all other measurements are correct:

$$\mathbf{b}_{\text{true}} = \begin{bmatrix} 22 \\ 34 \\ 29 \\ 31 \end{bmatrix}.$$

The percentage error in the faulty reading is $|21 - 22|/22 \times 100\% \approx 4.55\%$.

Determine the corrected amounts $\mathbf{x}_{\text{true}}$ and the absolute relative error $|x - x_{\text{true}}|/x_{\text{true}}$ for each compound.

(d) Error amplification (5 points)

Your results from part (c) show that some relative errors in the calculated amounts are much larger than the sensor's 4.55% error. Which concept from this course explains this behaviour? Briefly explain your reasoning.